

KIT AI Career Recommendation Hub

Project Summary & Resume Analysis Report

Project Purpose

AI Career Recommendation Hub is a Python desktop application built with CustomTkinter that evaluates a student's aptitude via a 15-factor psychometric slider assessment, then uses a trained RandomForestClassifier to predict the best-fit career paths and Google Gemini AI to generate personalized career analysis reports. The app also includes a full-featured AI Resume Builder with live preview and PDF export.

Architecture

main.py (1,559 lines) - Complete GUI, all pages, threading, PDF export
ml_engine.py (169 lines) - RandomForest training & prediction engine
ai_service.py (175 lines) - Gemini API integration with offline fallbacks
export_summary_pdf.py - This project summary PDF generator
career_model.pkl / scaler.pkl - Persisted trained model and feature scaler
.env - Environment file storing GEMINI_API_KEY

Technology Stack

GUI Framework:	CustomTkinter with dark GitHub-inspired theme (1400x900)
ML Model:	RandomForestClassifier (200 estimators, depth=15, 30 classes)
Training Data:	Synthetic - 200 samples x 30 careers = 6,000 total samples
AI Backend:	Google Gemini gemini-2.5-flash (structured JSON + text outputs)
PDF Export:	fpdf2 + matplotlib (embedded bar chart)
Assessment:	15 features: Math, Science, English, Aptitude Space, Technical Knack, Communication, Logical Reasoning, Creative Ability, Arts Interest, Science Interest, Commercial Awareness, Extroversion, Leadership, Problem Solving, Computer Science

Key Application Features

- * Login & Registration with in-memory user database
- * Dashboard sidebar navigation (Overview, Assessment, History, Resume Builder)
- * Neural Assessment: 15 sliders grouped in 3 cognitive categories
- * ML Pipeline: RandomForest inference returning Top-3 career matches with probability
- * Gemini AI: 5-paragraph personalized career analysis (salary, growth, skills)
- * Results Page: 3-column layout (Profile | Recommendation | Alternatives)
- * Results History: timestamped past assessment records per user

- * PDF Career Report: student details, ML bars, skill chart, AI analysis
- * Resume Builder: photo upload, dynamic education/experience rows, skill tags
- * Resume AI: Gemini-generated career objective, summary, key strengths (JSON)
- * Resume PDF Export: formatted A4 resume with colored header block
- * Full Offline Fallback: all AI functions work gracefully without Gemini API key

Student 1: Komal Priya - Resume Builder Analysis

Resume generated using the AI Resume Builder feature of the application

Personal Information

Name: Komal Priya
Email: komalpriya337@gmail.com
Phone: +91 6204929339
Location: Arwal, Bihar
Target Role: Marketing Professional
LinkedIn: linkedin.com/in/komal-priya-b5a69b2b3

AI-Generated Career Objective

To secure a dynamic Marketing position, leveraging practical experience in marketing and business development to drive impactful campaigns and contribute significantly to organizational growth.

AI-Generated Professional Summary

A results-oriented Master's graduate with nearly a year and a half of combined experience in Marketing and Business Development. Proven ability to engage clients, drive outreach, and contribute to business growth, acquired through roles at Inspior Consultant and Ai Expert Academy. Eager to apply a comprehensive understanding of marketing principles and practical skills to execute successful campaigns and achieve strategic objectives within a dynamic marketing team.

Key Strengths (AI-Generated)

- * Client Relationship Management
- * Strategic Business Development
- * Marketing Campaign Support
- * Effective Communication & Outreach

Education

- * Matriculation - Rajkiya Krit High School, Bharaoob (2016)
- * Intermediate - Rajkiya Krit High School, Bharaoob (2018)
- * Bachelor Degree - Sachinda Nand Shina College, Aurangabad, Bihar (2022)
- * Master's Degree - Gandhi Engineering College, BBSR, Odisha (2025)

Work / Internship Experience

- * Marketing Intern at Inspior Consultant (3 Months)
- * Business Development Manager at Ai Expert Academy, BBSR (11 Months)

Resume Quality Assessment

Strengths: Komal's resume effectively highlights her marketing and BDM experience. The AI-generated objective and summary are professional and role-specific. Good diversity of educational background from matriculation to Master's level.

Suggestions: Add quantifiable achievements (e.g., "increased outreach by 30%"). Expand the skills section with specific marketing tools (Google Ads, CRM, etc.). Consider adding a portfolio link or project descriptions for campaigns worked on.

Student 2: Rohit - KIT AI Career Profile Report

Career assessment result generated by the ML + Gemini pipeline

Student Details

Full Name:	Rohit
Username:	Rohit
College:	KIT (Kalam Institute of Technology)
Academic Stream:	Science

Optimal Career Matches (RandomForest ML Model)

#1 Sales Manager	33.6% Match	
#2 Cybersecurity Analyst	24.1% Match	
#3 HR Specialist	11.4% Match	

Notable Assessment Scores

Computer Science:	100/100 (Outstanding)
Problem Solving:	96/100 (Exceptional)
Leadership:	89/100 (Strong)
Extroversion:	81/100 (High)
Commercial Awareness:	75/100 (Good)
Communication:	30/100 (Needs Improvement)
English:	16/100 (Needs Improvement)

Gemini AI Career Analysis

Your exceptional Problem Solving (96), Leadership (89), Extroversion (81), and perfect Computer Science (100) score make Sales Manager a highly aligned career. Your Commercial Awareness (75) adds valuable market intuition for a sales leadership role.

Development Areas: Communication (30) and English (16) are critical gaps. Persuasive communication is at the core of successful sales management. Focused improvement in these areas will significantly unlock your potential.

Required Skills for Sales Manager

- * Advanced Communication and Negotiation Skills
- * Strategic Leadership and Team Building
- * Data Analysis and CRM Proficiency
- * Problem Solving and Adaptability in dynamic markets

Market Insights (AI)

Salary Range:	\$70,000 - \$150,000 per year + commission / bonus structures
Growth Trends:	Stable to slightly increasing demand over the next decade. Sales Managers who can leverage data analytics, CRM tools, and digital channels will be in highest demand globally.

Student 3: Mohit - KIT AI Career Profile Report

Career assessment result generated by the ML + Gemini pipeline

Student Details

Full Name:	Mohit
Username:	Mohit
College:	Kalam Institute of Technology (KIT)
Academic Stream:	Science

Optimal Career Matches (RandomForest ML Model)

#1 Product Manager	16.1% Match	
#2 Cybersecurity Analyst	14.9% Match	
#3 Sales Manager	9.5% Match	

Notable Assessment Scores

Computer Science:	100/100 (Perfect)
English:	100/100 (Perfect)
Extroversion:	95/100 (Exceptional)
Leadership:	90/100 (Strong)
Commercial Awareness:	80/100 (Strong)
Problem Solving:	75/100 (Good)
Creative Ability:	45/100 (Needs Development)
Logical Reasoning:	50/100 (Average)

Gemini AI Career Analysis

Perfect scores in Computer Science (100) and English (100), combined with outstanding Extroversion (95) and Leadership (90), make Product Manager a natural career fit. Strong Commercial Awareness (80) and Science Interest position you well for understanding market needs and guiding development teams toward impactful product solutions.

Development Areas: Creative Ability (45) could be cultivated for better feature ideation and user-centered thinking. Strengthening Logical Reasoning (50) will help in dissecting complex product problems and formulating robust product-market strategies.

Required Skills for Product Manager

- * Product Vision and Strategy - roadmap definition and prioritization
- * Stakeholder Management and Cross-functional Leadership

- * Technical Understanding for effective engineering communication
- * User Empathy and Market Research for impactful product decisions

Market Insights (AI)

Salary Range:	\$80,000 - \$180,000+ annually for experienced professionals
Growth Trends:	Exceptionally strong. As digital product development accelerates, demand for skilled Product Managers is projected to grow significantly over the next decade across all sectors.

Code Analysis & Improvement Suggestions

Technical review of the project codebase quality

Code Quality Overview

main.py (1,559 lines):	GUI + all pages. Well-structured but monolithic.
ml_engine.py (169 lines):	Clean, well-documented. Excellent auto-retrain logic.
ai_service.py (175 lines):	Best module. Typed, documented, fallback, JSON parsing.
Overall Architecture:	Good 3-layer separation: GUI / ML / AI Service.

[!] Critical Issues

[!] In-Memory User Database

All registered users and history are lost every time the app is closed. Fix: Persist to SQLite (built-in) or a JSON file.

[!] Plaintext Password Storage

Passwords stored as raw strings in a Python dict. Fix: Hash with hashlib.sha256 before storing and compare hashes.

[!] False Model Accuracy Display

The "94.6% accuracy" shown on the dashboard is training accuracy, not real accuracy. Fix: Use sklearn train_test_split and evaluate on a held-out test set.

[~] Moderate Issues

- * import statements inside methods (slow, non-Pythonic) - move all to top of file.
- * MLPredictor initialized at module level before GUI starts - can cause silent startup freeze.
- * Interests section (Science High/Commerce Medium/Arts Low) is hard-coded, never changes.
- * Default login credentials (Rohit / 1234) pre-filled in UI - remove before deployment.
- * Two code sections are both labeled "#4" in comments - numbering error.

[+] Recommended Improvements

- * Persist users and history to SQLite using built-in sqlite3 module (no extra dependency).
- * Hash passwords with hashlib.sha256 before storing.
- * Use train_test_split to evaluate model on unseen data before showing accuracy.
- * Add a startup splash screen while model loads to prevent apparent freeze.
- * Break main.py into page modules: login_page.py, results_page.py, resume_page.py.
- * Add unit tests for ml_engine.predict() and ai_service fallback paths.
- * Replace bare except: clauses with except Exception: to avoid swallowing signals.
- * Use named FEATURES index constants instead of magic numbers like original_scores[6].
- * Add input sanitization for registration fields (length, special characters).

* Cache Gemini API responses locally to reduce redundant API calls.